

Assignment : A+- Troubleshooting and helpdesk

Day 3: BIOS and POST Troubleshooting Theory (30 min): Understanding BIOS and POST errors. Practical (30 min): Analyze POST error codes on a given setup

TASK : 1

List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.



No.	BIOS/POST Errors	Usually Indicate
1.	CMOS Checksum Error	CMOS battery failure or corrupted
2	Keyboard Failure	Keyboard not detected or loose connection or port failure
3	Memory Failure	RAM may be loose
4	No boot device found	HDD/SSD may not detects
5	CPU Fan	May fan not rotating or physically damaged

When a PC starts,

Power on → BIOS → POST → Hardware check → Boot Device → Operating System

CMOS Checksum Error : This error can occur when Expected BIOS/CMOS settings do not match expected checksum. Due to faulty or damaged battery of cmos.

TASK : 2

Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.



Problem:

When the PC is turned on, it produces 3 beeps in a repeating pattern and there is no display on the monitor.

Meaning:

For a traditional AMI BIOS, 3 beeps generally indicate a main memory (RAM) read/write test error during POST. This means that the BIOS is having a problem detecting or testing the RAM.

Troubleshooting Steps:

1. **Reseat the RAM:** Turn off the computer, disconnect the power, remove the RAM, check it for dust or damage, and install it firmly again.
2. **Test RAM individually:** If more than one RAM module is installed, test one module at a time and try different compatible RAM slots. This can help identify a faulty RAM module or RAM slot.

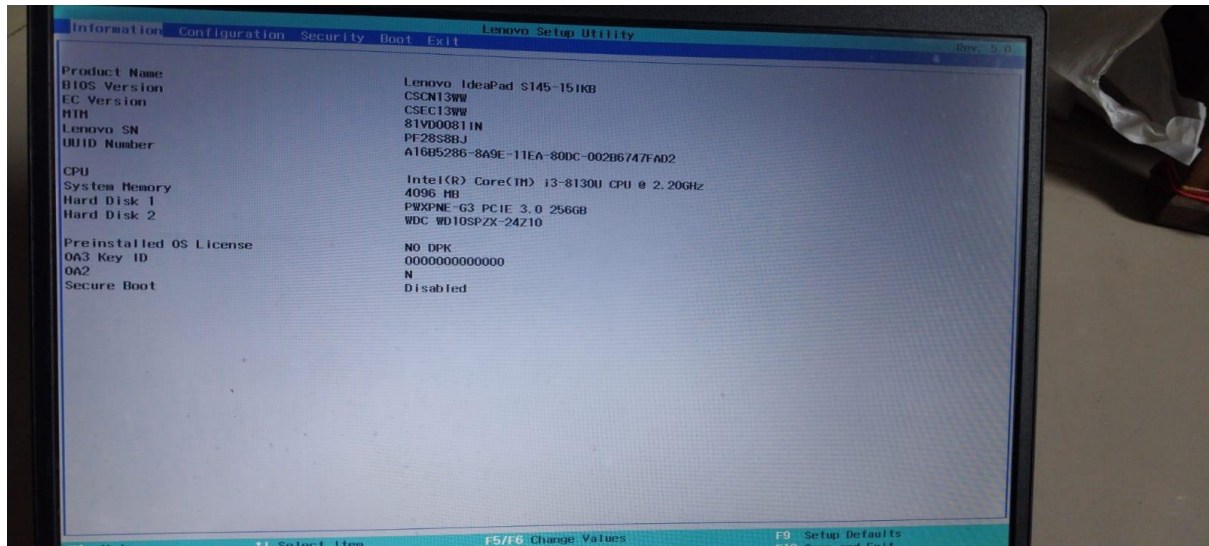
Conclusion:

Three repeating beeps with no display can indicate a RAM-related POST problem. Reseating and testing the RAM modules individually are two basic troubleshooting steps that can help identify the problem.

3 Beeps → RAM Error → Remove RAM → Clean and RESEAT RAM → Test one RAM at a time

TASK : 3

Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



BIOS VERSION CSCN13WW

MTM 81VD0081IN

CPU Intel(R) Core(TM) i3-8130U CPU@ 2.20GHz

System Memory 4096 MB

Hard Disk 1 PWXPNE –G3 PCIE 3.0 256GB

TASK : 4

You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.
Hint: Think about the CMOS battery and BIOS settings reset



The message “CMOS checksum error – Defaults loaded” means that the BIOS has detected that the stored CMOS settings are invalid or have changed unexpectedly. As a result, the BIOS loads its default settings to allow the computer to start.

Common Causes:

- Weak or dead CMOS/RTC battery
- Corrupted or reset BIOS settings
- Incorrect BIOS configuration
- BIOS-related problems

Resolution:

1. Restart the computer and enter the BIOS/UEFI setup.
2. Check the date and time. If they reset repeatedly, the CMOS battery may be weak.
3. Check and configure the required BIOS settings, including the correct boot device.
4. Save the settings and restart the computer.
5. If the error continues, replace the CMOS/RTC battery with a compatible battery.
6. Start the computer again and check whether the settings are retained.

Conclusion:

A CMOS checksum error usually means that the BIOS has detected invalid or changed CMOS settings. A weak CMOS battery is one common cause. Reconfiguring the BIOS settings and replacing the battery when necessary can resolve the problem.